

Module 3: Data Link Layer and Wireless Networks



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Data-Link Layer

- Data Link Control (DLC)
- Multiple Access Protocols (MAC)
- Link-Layer Addressing
- Ethernet Protocol
- Connecting Devices

Wireless Networks

- Wireless LANs
- Mobile IP

Hands-On

- Datalink Provider Interface
- SOCK_PACKET
- PF_PACKET

5.3 Multiple Access Protocols

- The Data-Link Layer has two sublayers:
 - **Data Link Control (DLC)**
 - **Media Access Control (MAC)**
- **Dedicated link:**
 - Only DLC protocol is required.
 - Example: PPP on a point-to-point link.
- **Shared medium:**
 - Multiple users share the same wire or wireless medium.
 - A MAC protocol is required to control access.

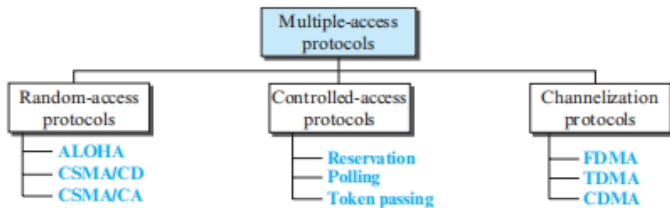
Main Question

Who gets permission to use the shared medium, and when?

Why Do We Need Multiple Access Protocols?

- Used when stations share a **multipoint or broadcast link**.
- Similar to rules for speaking in an assembly.
- Protocol should ensure:
 - Only one station transmits at a time.
 - Stations do not interrupt each other.
 - No station monopolizes the medium.
 - Every station gets a fair chance.
- Main goals:
 - 1 Prevent collisions.
 - 2 Handle collisions if they occur.

Figure 5.28 *Taxonomy of multiple-access protocols discussed in this chapter*



Categories of Multiple Access Protocols

Category	Main Idea
Random Access	Stations compete for the shared medium.
Controlled Access	Stations coordinate before transmission.
Channelization	Bandwidth is divided by time, frequency, or code.

Protocols Discussed

- **Random Access:** ALOHA, CSMA, CSMA/CD, CSMA/CA
- **Controlled Access:** Reservation, Polling, Token Passing
- **Channelization:** Time, Frequency, and Code division

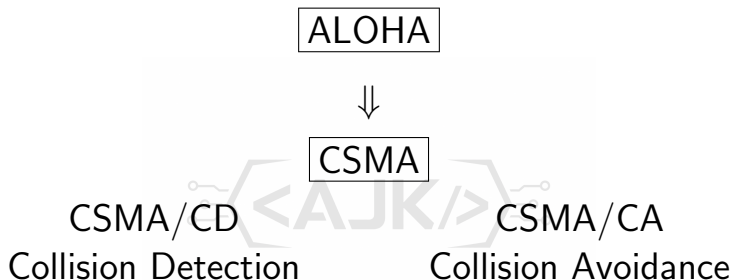
5.3.1 Random Access

- Also called **contention methods**.
- No station is superior to another.
- No central controller determines who transmits next.
- Stations compete for access to the shared medium.
- A station decides whether to transmit based on:
 - Medium is **idle**
 - Medium is **busy**

Four Important Questions

- 1 When can a station access the medium?
- 2 What happens if the medium is busy?
- 3 How is success or failure determined?
- 4 What happens after a collision?

Evolution of Random Access Protocols



- **ALOHA:** Transmit whenever ready.
- **CSMA:** Sense the medium before transmitting.
- **CSMA/CD:** Detect and handle collisions.
- **CSMA/CA:** Try to avoid collisions.

ALOHA: Introduction

- Earliest random-access method.
- Developed at the **University of Hawaii** in the early 1970s.
- Originally designed for wireless LANs.
- Can be used on any shared medium.
- Main problem: **Collisions**.

ALOHA Principle

Stations compete for the same shared channel.

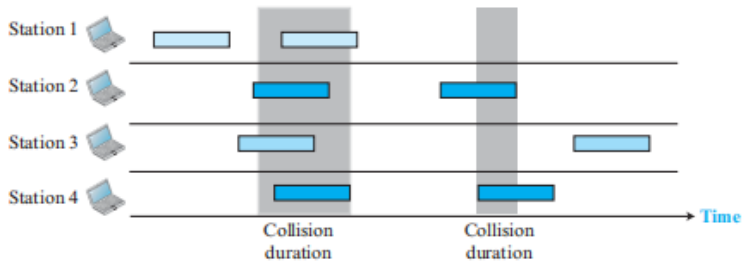
- If two stations transmit simultaneously:
 - Frames collide.
 - Frames become garbled.
 - Retransmission is required.

Basic Rule

Send whenever you have a frame to send.

- No synchronization between stations.
- Multiple stations can transmit at any time.
- Even one overlapping bit causes a collision.
- Collided frames are destroyed.

Figure 5.29 *Frames in a pure ALOHA network*



Pure ALOHA: Retransmission

- ① Station sends a frame.
- ② Receiver should send an acknowledgment.
- ③ If ACK does not arrive before timeout:
 - Frame or ACK is assumed destroyed.
 - Station retransmits the frame.

Problem

If all collided stations retransmit immediately, another collision may occur.

Solution: Random Back-off

Wait for random time T_B before retransmitting

Pure ALOHA: Timeout and Back-off

- Maximum timeout:

$$T_{\text{timeout}} = 2T_p$$

- T_p = maximum propagation time.
- Binary exponential back-off:

- Back-off time:

$$R = 0 \text{ to } 2^K - 1$$

$$T_B = R \times T_p$$

or

$$T_B = R \times T_{fr}$$

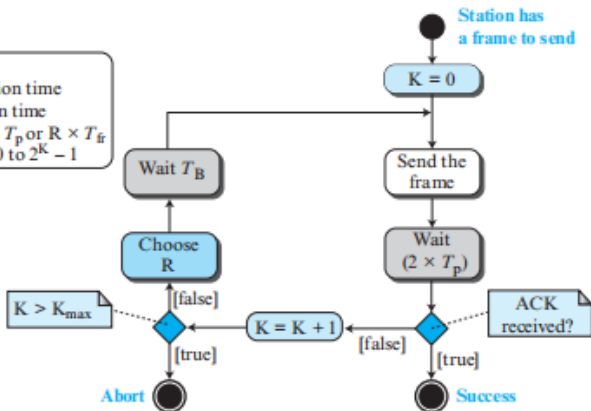
- K = number of unsuccessful attempts.
- Usually:

$$K_{\text{max}} = 15$$

Figure 5.30 Procedure for pure ALOHA protocol

Legend

K : Number of attempts
 T_p : Maximum propagation time
 T_{fr} : Average transmission time
 T_B : (Back-off time): $R \times T_p$ or $R \times T_{fr}$
 R : (Random number): 0 to $2^K - 1$



Example: Pure ALOHA Back-off

Given

Maximum distance = 600 km

Signal speed:

$$3 \times 10^8 \text{ m/s}$$

$$T_p = \frac{600 \times 10^3}{3 \times 10^8} = 2 \text{ ms}$$

For $K = 2$:

$$R = 0, 1, 2, 3$$

Therefore:

$$T_B = R \times 2 \text{ ms}$$

Possible Back-off Times

$$0, 2, 4, 6 \text{ ms}$$

Pure ALOHA: Vulnerable Time

- Assume frame transmission time:

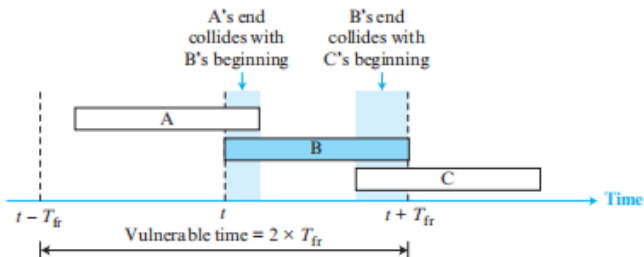
$$T_{fr}$$

- Collision can occur if another station starts:
 - Up to T_{fr} before transmission begins.
 - Up to T_{fr} after transmission begins.

Pure ALOHA Vulnerable Time

$$2T_{fr}$$

Figure 5.31 Vulnerable time for pure ALOHA protocol



Example: Vulnerable Time

Given

Frame size = 200 bits

Channel bandwidth = 200 kbps

$$T_{fr} = \frac{200}{200 \text{ kbps}} = 1 \text{ ms}$$

Therefore:

$$\text{Vulnerable Time} = 2T_{fr} = 2 \text{ ms}$$

- No station should start transmission:
 - 1 ms before this station starts.
 - During the 1 ms transmission period.

Pure ALOHA: Throughput

Let:

G = Average frames generated during one T_{fr}

Throughput

$$S = Ge^{-2G}$$

Maximum throughput:

$$S_{\max} = \frac{1}{2e} = 0.184$$

when:

$$G = \frac{1}{2}$$

Maximum efficiency of Pure ALOHA = 18.4%

Pure ALOHA: Throughput Example

Given:

$$T_{fr} = 1 \text{ ms}$$

Generated Frames	G	$S = Ge^{-2G}$	Successful Frames
1000 frames/s	1	0.135	$1000(0.135) = 135$
500 frames/s	0.5	0.184	$500(0.184) = 92$
250 frames/s	0.25	0.152	$250(0.152) = 38$

Observation

The maximum percentage throughput occurs when:

$$G = 0.5$$

Slotted ALOHA

- Developed to improve Pure ALOHA.
- Time is divided into slots.
- Each slot duration:

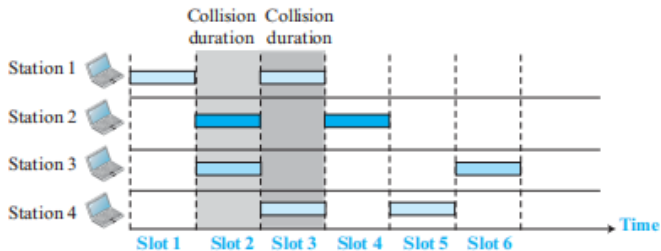
$$T_{fr}$$

- Stations can transmit **only at the beginning of a slot.**
- If a station misses a slot:
 - It must wait for the next slot.

Important

Collisions can occur only when two or more stations transmit at the beginning of the same slot.

Figure 5.32 *Frames in a slotted ALOHA network*



Slotted ALOHA: Vulnerable Time and Throughput

Vulnerable Time

$$T_{fr}$$

- Vulnerable time is half of Pure ALOHA.

Throughput

$$S = Ge^{-G}$$

Maximum throughput:

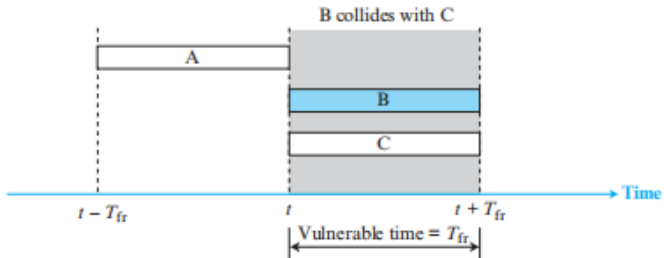
$$S_{\max} = \frac{1}{e} = 0.368$$

when:

$$G = 1$$

Maximum efficiency = 36.8%

Figure 5.33 *Vulnerable time for slotted ALOHA protocol*



Pure ALOHA vs Slotted ALOHA

	Pure ALOHA	Slotted ALOHA
Transmission	Anytime	Beginning of slot only
Synchronization	Not required	Required
Vulnerable Time	$2T_{fr}$	T_{fr}
Throughput	Ge^{-2G}	Ge^{-G}
Maximum S	0.184	0.368
Maximum Efficiency	18.4%	36.8%
Maximum at	$G = 0.5$	$G = 1$

Conclusion

Slotted ALOHA doubles the maximum efficiency of Pure ALOHA.

Carrier Sense Multiple Access (CSMA)

Basic Principle

Sense Before Transmit / Listen Before Talk

- Station checks the medium before transmission.
- If medium is idle:
 - Station may transmit.
- If medium is busy:
 - Station follows a persistence procedure.

Advantage

Reduces the probability of collision compared with ALOHA.

Limitation

CSMA cannot completely eliminate collisions.

Why Can Collisions Still Occur in CSMA?

- Because of **propagation delay**.
- A station may sense the channel as idle because:
 - The signal from another station has not yet reached it.
- Example:
 - Station B senses idle and starts transmission at t_1 .
 - Station C senses idle at t_2 .
 - B's signal has not yet reached C.
 - C also starts transmission.
 - Collision occurs.

Insert Textbook Figure 5.34: Collision Due to Propagation Delay

CSMA Vulnerable Time

$$T_p$$

- T_p = propagation time.
- Collision can occur if another station transmits during this period.
- Once the first bit reaches all stations:
 - Other stations sense the medium as busy.
 - They refrain from transmitting.

Insert Textbook Figure 5.35: Vulnerable Time in CSMA

CSMA Persistence Methods

When a station wants to send:

- What if the channel is **idle**?
- What if the channel is **busy**?

Three Persistence Methods

- 1 **1-Persistent**
- 2 **Nonpersistent**
- 3 **p -Persistent**

Figure 5.36 Behavior of three persistence methods

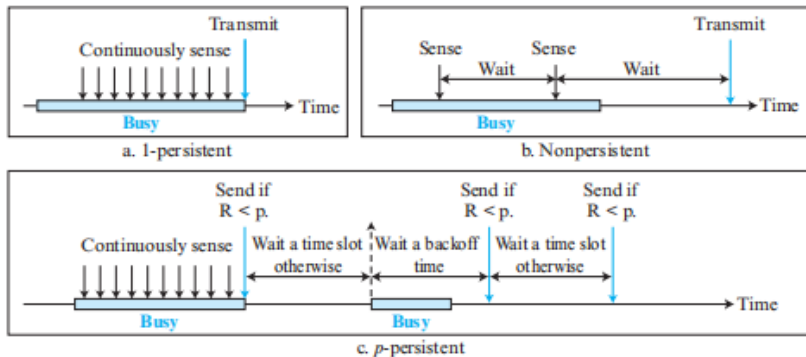
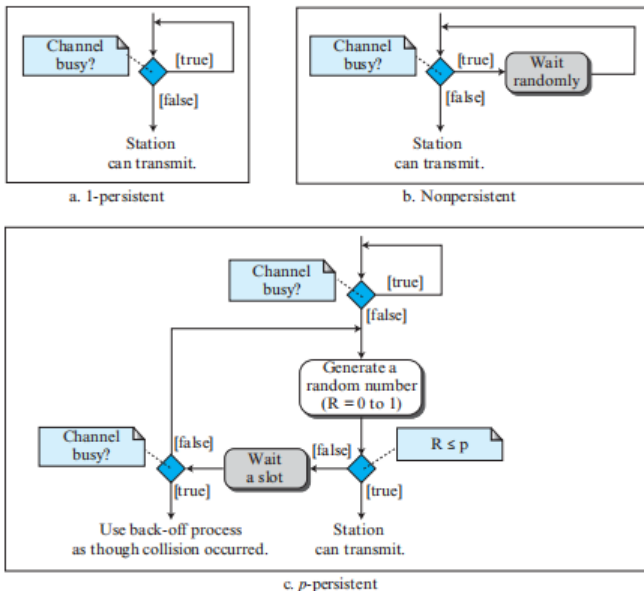


Figure 5.37 Flow diagram for three persistence methods



1-Persistent and Nonpersistent CSMA

Feature	1-Persistent	Nonpersistent
If Idle	Send immediately	Send immediately
If Busy	Keep sensing	Wait random time
Transmission Probability	1	–
Collision Chance	High	Lower
Efficiency	Higher utilization	Medium may remain idle

Key Idea

- **1-Persistent:** Aggressive approach.
- **Nonpersistent:** Random waiting reduces collisions.

p -Persistent CSMA

Used when the channel has synchronized time slots.

After Finding the Channel Idle

- 1 With probability p :

Transmit

- 2 With probability:

$$q = 1 - p$$

wait until the next time slot.

Then:

- If channel remains idle $\rightarrow$ repeat decision.
- If channel becomes busy $\rightarrow$ use back-off procedure.

Advantage

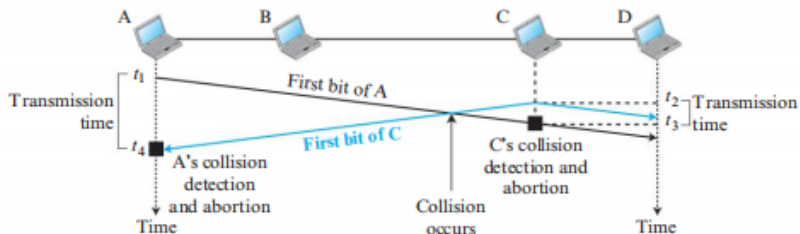
Balances:

Collision Reduction + Channel Efficiency

Carrier Sense Multiple Access with Collision Detection

- Station senses the medium before transmission.
- While transmitting, station continuously monitors the medium.
- If no collision:
 - Transmission completes successfully.
- If collision is detected:
 - Abort transmission.
 - Send a jamming signal.
 - Wait using back-off.
 - Retransmit.

Figure 5.38 Collision of the first bits in CSMA/CD



CSMA/CD: Continuous Monitoring

Main Difference from ALOHA

- **ALOHA:**

Transmit entire frame → Wait for ACK

- **CSMA/CD:**

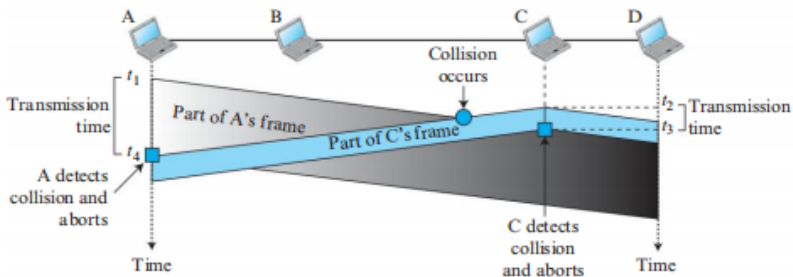


Transmit + Monitor simultaneously

Transmission Stops When

- 1 Entire frame transmission is complete, OR
- 2 Collision is detected.

Figure 5.39 Collision and abortion in CSMA/CD



CSMA/CD: Minimum Frame Size

For collision detection:

Requirement

The sender must still be transmitting when a collision signal returns.

Worst case:

$$T_{fr} \geq 2T_p$$

where:

- T_{fr} = frame transmission time.
- T_p = maximum propagation time.

Reason

- Signal takes T_p to reach the farthest station.
- Collision effect takes another T_p to return.

Therefore:

$$\text{Minimum transmission time} = 2T_p$$

Example: Minimum Frame Size

Given

Bandwidth:

10 Mbps

Maximum propagation time:

$$T_p = 25.6 \mu s$$

$$T_{fr} = 2T_p = 51.2 \mu s$$

Minimum frame size:

$$10 \text{ Mbps} \times 51.2 \mu s$$

$$512 \text{ bits} = 64 \text{ bytes}$$

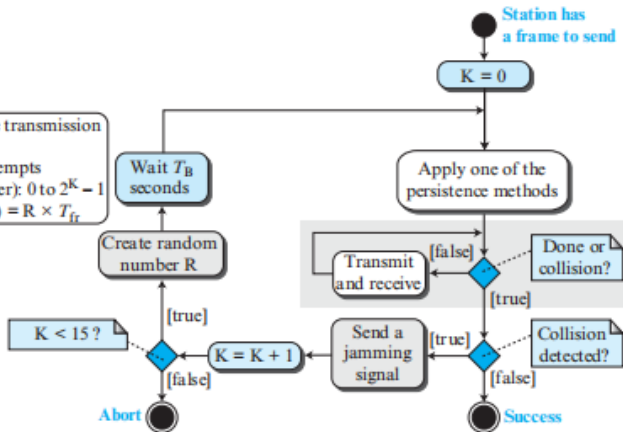
CSMA/CD Procedure

- ❶ Sense the channel.
- ❷ Apply a persistence method.
- ❸ Start transmission when allowed.
- ❹ Continuously monitor the channel.
- ❺ If transmission completes:
 - Success.
- ❻ If collision occurs:
 - Abort transmission.
 - Send jamming signal.
 - Wait using back-off.
 - Retry if attempts are below maximum.

Figure 5.40 Flow diagram for the CSMA/CD

Legend

T_{fr} : Frame average transmission time
 K : Number of attempts
 R : (random number): 0 to $2^K - 1$
 T_B : (Back-off time) = $R \times T_{fr}$



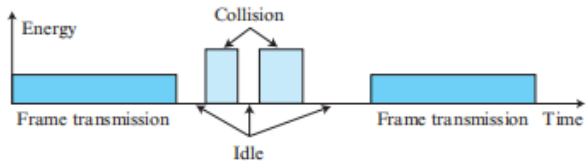
Energy Level in CSMA/CD

The energy level in the channel can have three states:

Energy Level	Meaning
Zero	Channel is idle
Normal	One station is transmitting successfully
Abnormal	Collision occurs; energy approximately twice normal

- Stations monitor energy to determine:
 - Idle channel
 - Busy channel
 - Collision

Figure 5.41 *Energy level during transmission, idleness, or collision*



CSMA/CD Throughput and Ethernet

- CSMA/CD throughput is greater than ALOHA.

Method	Maximum Throughput
1-Persistent CSMA/CD	Around 50% when $G = 1$
Nonpersistent CSMA/CD	Up to 90% when G is between 3 and 8

Traditional Ethernet

- Used CSMA/CD.
- Data rate: 10 Mbps.
- Broadcast LAN.
- Used 1-persistent access method.

Carrier Sense Multiple Access with Collision Avoidance

- Variation of CSMA.
- Attempts to **avoid collisions**.
- Used in **wireless LANs**.
- Discussed in detail with wireless LAN protocols.

Quick Comparison

CSMA/CD: Detect Collision

CSMA/CA: Avoid Collision

5.3.2 Controlled Access

- Stations consult one another before transmission.
- A station cannot transmit without authorization.
- Access is controlled to avoid collisions.

Three Controlled Access Methods

- 1 Reservation
- 2 Polling
- 3 Token Passing

Reservation Method

- A station reserves the medium before transmitting.
- Time is divided into intervals.
- Each interval contains:
 - 1 Reservation frame
 - 2 Data frames

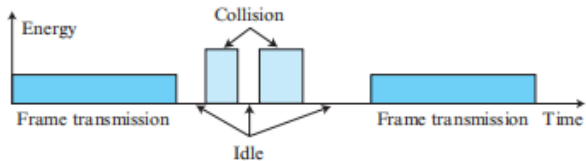
If there are N stations:



N reservation minislots

- Each station owns one minislot.
- Station makes reservation in its own minislot.
- Reserved stations transmit data afterward.

Figure 5.41 *Energy level during transmission, idleness, or collision*



Polling

- One device acts as **primary**.
- Other devices act as **secondary**.
- Primary controls the link.
- Secondary stations follow instructions.
- Primary initiates communication.

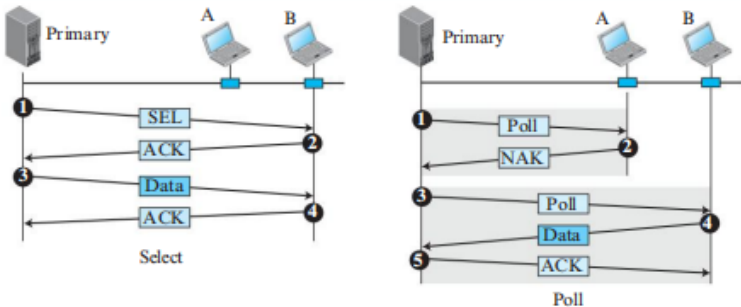
Two Main Functions

- 1 Select
- 2 Poll

Main Drawback

If the primary station fails, the entire system can fail.

Figure 5.43 *Select and poll functions in polling-access method*



Polling: Select Function

Used When the Primary Wants to Send Data

- 1 Primary has data to send.
- 2 Primary knows the link is available.
- 3 Primary does not know whether target is ready.
- 4 Primary sends a **Select (SEL) frame**.
- 5 SEL frame contains target secondary's address.
- 6 Primary waits for ready acknowledgment.
- 7 Data transmission begins.

Simple Idea

Primary asks: “Are you ready to receive?”

Used When the Primary Wants to Receive Data

- ① Primary polls secondary stations one by one.
- ② Secondary responds:
 - **NAK:** No data to send.
 - **Data frame:** Has data to send.
- ③ If NAK:
 - Primary polls the next secondary.
- ④ If data frame:
 - Primary receives data.
 - Primary sends ACK.

Token Passing

- Stations form a **logical ring**.
- Each station has:
 - Predecessor
 - Successor
- A special packet called a **token** circulates.
- Possession of the token gives the right to transmit.

Rule

No Token $\Rightarrow$ No Transmission

Token $\Rightarrow$ Right to Access Medium

Token Passing Procedure

- ① Station waits for the token.
- ② Station receives token.
- ③ If it has data:
 - Holds token.
 - Sends data.
- ④ After transmission:
 - Releases token.
 - Passes token to successor.
- ⑤ If station has no data:
 - Immediately passes token.



Result

Only the station holding the token can transmit.

Token Management

Token passing requires management to ensure reliable operation.

- Limit the time a station can hold the token.
- Detect a lost or destroyed token.
- Handle station failure while holding the token.
- Assign priorities to:
 - Stations
 - Types of data
- Allow high-priority stations to obtain access.
- Force low-priority stations to release the token.

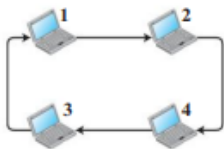
Logical Ring

- Token passing does **not** require a physical ring.
- Different physical topologies can create a logical ring.

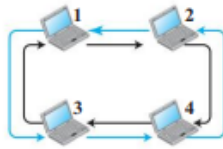
Physical Topologies Supporting a Logical Ring

- 1 Physical Ring
- 2 Dual Ring
- 3 Bus Ring (Token Bus)
- 4 Star Ring

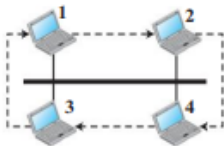
Figure 5.44 *Logical ring and physical topology in token-passing access method*



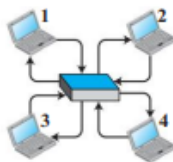
a. Physical ring



b. Dual ring



c. Bus ring



d. Star ring

Physical Ring and Dual Ring

Physical Ring

- Stations physically connected in a ring.
- Token moves to successor.
- Token does not need successor address.
- Major drawback:
 - Failure of one link can fail the whole system.

Dual Ring

- Second ring operates in reverse direction.
- Used for emergencies.
- If main ring fails:
 - Rings combine temporarily.
- Used by:
 - FDDI
 - CDDI
- Requires:
 - Two transmitter ports
 - Two receiver ports

Bus Ring and Star Ring

Bus Ring (Token Bus)

- Physical topology: Bus.
- Logical topology: Ring.
- Each station knows:
 - Successor address
 - Predecessor address
- Token contains successor address.
- Used by IEEE Token Bus LAN.

Star Ring

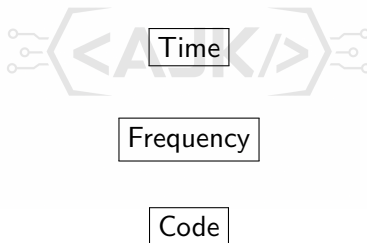
- Physical topology: Star.
- Hub creates the logical ring.
- Failed link can be bypassed.
- Adding/removing stations is easier.
- Used in IBM Token Ring LAN.

5.3.3 Channelization

Definition

Channelization, also called **channel partition**, divides the available bandwidth among stations.

Bandwidth can be shared using:



- Commonly used in wireless networks.
- Detailed discussion belongs to wireless networking.

Multiple Access Protocols: Summary

Method	Key Idea
Pure ALOHA	Send anytime; vulnerable time $2T_{fr}$
Slotted ALOHA	Send only at slot boundaries; vulnerable time T_{fr}
CSMA	Sense medium before transmission
CSMA/CD	Sense + detect collision + abort transmission
CSMA/CA	Sense + attempt to avoid collision
Reservation	Reserve a slot before transmitting
Polling	Primary controls secondary stations
Token Passing	Token holder gets transmission right
Channelization	Divide medium by time, frequency, or code

Important Formula Sheet

Pure ALOHA

$$\text{Vulnerable Time} = 2T_{fr}$$

$$S = Ge^{-2G}$$

$$S_{\max} = 0.184 \quad \text{at } G = 0.5$$

Slotted ALOHA

$$\text{Vulnerable Time} = T_{fr}$$

$$S = Ge^{-G}$$

$$S_{\max} = 0.368 \quad \text{at } G = 1$$

CSMA/CD

$$T_{fr} \geq 2T_p$$

Key Takeaways

- ① Shared media requires controlled access.
- ② Random access allows stations to compete.
- ③ ALOHA is simple but has high collision probability.
- ④ Slotted ALOHA improves efficiency through synchronization.
- ⑤ CSMA uses the principle:

Listen Before Talk

- ⑥ CSMA/CD detects collisions while transmitting.
- ⑦ Controlled access prevents collisions through coordination.
- ⑧ Token passing gives access rights through a circulating token.
- ⑨ Channelization divides bandwidth among users.